

Student Grade Calculator

1. Project Overview

The Student Grade Calculator is a Python console-based application developed as part of Phase 1 – Module 1. The application demonstrates the use of Python fundamentals such as variables, data types, input/output, control structures, loops, functions, error handling, and string formatting to calculate and display student grade reports.

2. Problem Statement

Managing student grades manually can be time-consuming and prone to calculation errors. This project automates the process by collecting student information, validating marks, calculating grades and performance statistics, and generating a structured report using Python.

3. Features

Input Management

- Student details collection
- Subject names entry
- Marks validation

Performance Analysis

- Total marks calculation
- Average marks calculation
- Highest and lowest marks identification
- Grade distribution analysis

Report Generation

- Subject-wise report
- Scholarship eligibility status
- Appreciation remarks
- Formatted student performance report

4. Technologies Used

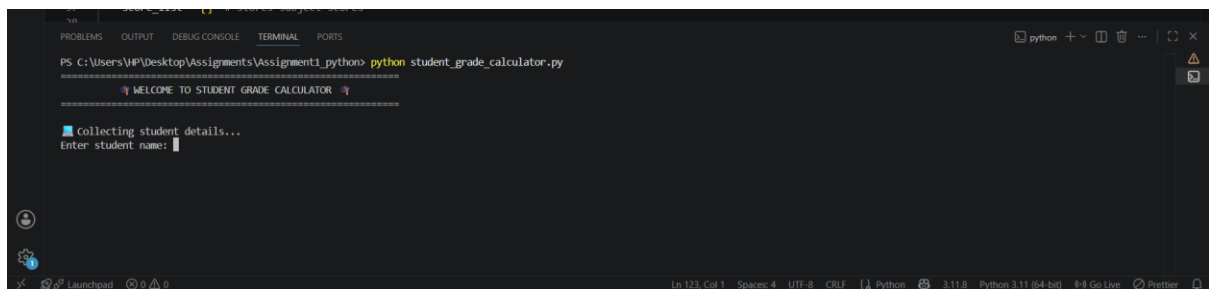
- Python 3
- Visual Studio Code

5. Program Workflow

- Start
- Display Welcome Message
- Collect Student Details (Name, Student ID, Number of Subjects)
- Validate Inputs
- Collect Subject Names and Marks
- Validate Marks (0–100)
- Calculate Results (Total, Average, Highest, Lowest, Grade, Pass Count)
- Generate Student Performance Report
- Calculate Grades for Another Student?
 - **Yes** → Return to Collect Student Details
 - **No** → Display Thank You Message
- End

6. Output Screenshots and Explanation

Welcome Screen



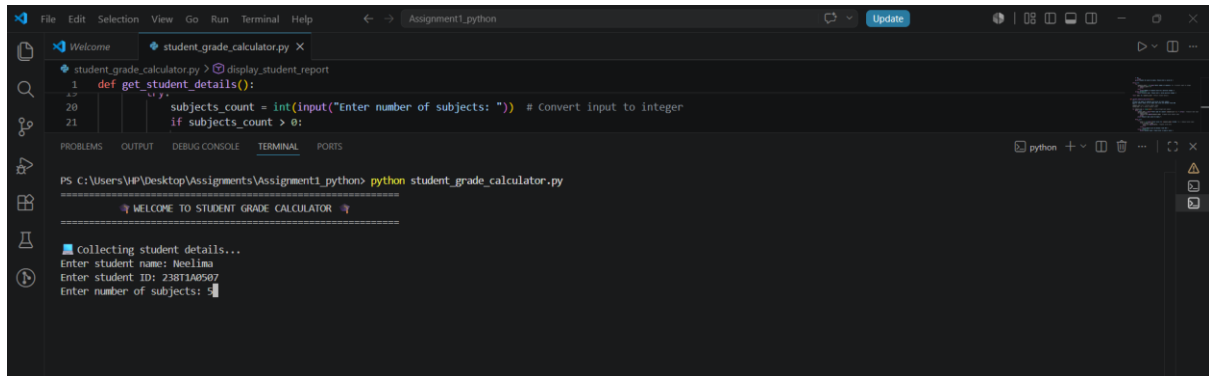
```
PS C:\Users\VP\Desktop\Assignments\Assignment1_python> python student_grade_calculator.py
=====
WELCOME TO STUDENT GRADE CALCULATOR
=====

collecting student details...
Enter student name: 
```

Explanation:

The application starts by displaying a welcome message. It introduces the Student Grade Calculator and indicates that the program is ready to collect student information.

Student Details Input

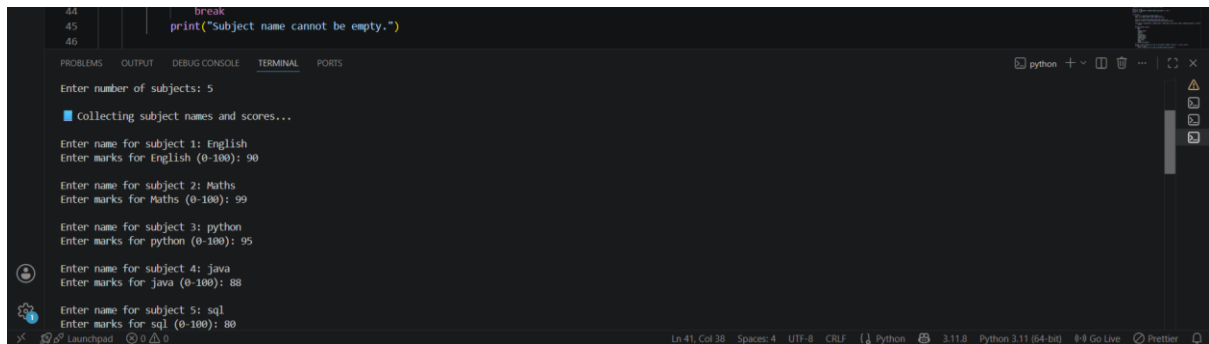


```
File Edit Selection View Go Run Terminal Help
student_grade_calculator.py
1 def get_student_details():
2     subjects_count = int(input("Enter number of subjects: ")) # Convert input to integer
3     if subjects_count > 0:
4         # Collecting student details...
5         Enter student name: Neelima
6         Enter student ID: 238TJA0507
7         Enter number of subjects: 5
```

Explanation:

The application collects the student's basic information. Input validation ensures that the student's name and ID are not empty and that the number of subjects is a positive integer.

Subject Marks Entry

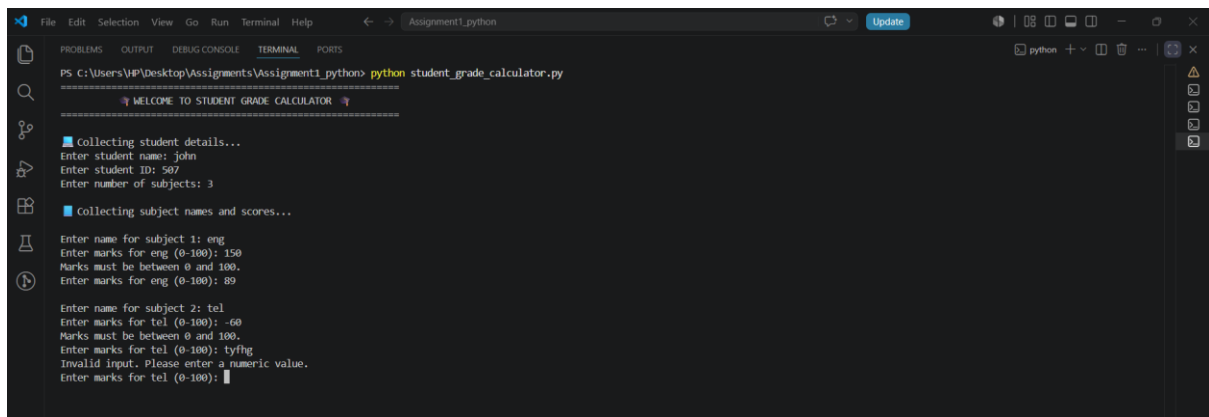


```
44 break
45 print("Subject name cannot be empty.")
46
Enter number of subjects: 5
collecting subject names and scores...
Enter name for subject 1: English
Enter marks for English (0-100): 90
Enter name for subject 2: Maths
Enter marks for Maths (0-100): 99
Enter name for subject 3: python
Enter marks for python (0-100): 95
Enter name for subject 4: java
Enter marks for java (0-100): 88
Enter name for subject 5: sql
Enter marks for sql (0-100): 80
```

Explanation:

The user enters each subject name and its corresponding marks. The application validates that marks are within the range of 0–100 and accepts only valid numeric input.

Input Validation



```
PS C:\Users\HP\Desktop\Assignments\Assignment1_python> python student_grade_calculator.py

=====
WELCOME TO STUDENT GRADE CALCULATOR
=====

collecting student details...
Enter student name: john
Enter student ID: 507
Enter number of subjects: 3

collecting subject names and scores...

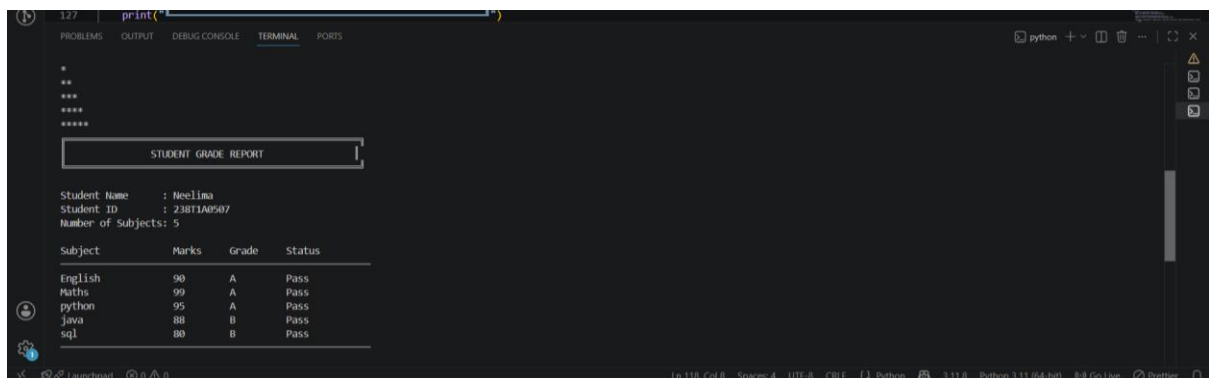
Enter name for subject 1: eng
Enter marks for eng (0-100): 150
Marks must be between 0 and 100.
Enter marks for eng (0-100): 89

Enter name for subject 2: tel
Enter marks for tel (0-100): -60
Marks must be between 0 and 100.
Enter marks for tel (0-100): tyfng
Invalid input. Please enter a numeric value.
Enter marks for tel (0-100):
```

Explanation:

The application uses exception handling and input validation to detect invalid entries. Appropriate error messages are displayed, prompting the user to provide valid input without terminating the program.

Generated Report



```
*****
*****
*****
*****
*****

STUDENT GRADE REPORT

Student Name      : Neelima
Student ID       : 23BIT1A09507
Number of Subjects: 5

Subject  Marks  Grade  Status
-----
English   90      A     Pass
Maths     99      A     Pass
python    95      A     Pass
java      88      B     Pass
sql       80      B     Pass
```

Explanation:

The generated report displays the student's information, subject-wise marks, grades, pass/fail status, and other academic details in a well-formatted table using formatted strings.

- Student details
- Subject-wise marks
- Grade for each subject
- Pass/Fail status

The information is displayed in a neatly formatted table using formatted strings.

Statistical Summary



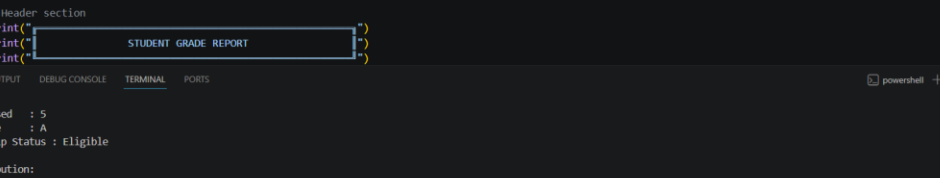
```
print("
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python + - |  X
Total Marks : 452
Average : 90.40
Highest Marks : 99 (Maths)
Lowest Marks : 80 (sql)
Subjects Passed : 5
Overall Grade : A
Scholarship Status : Eligible
Grade Distribution:
A : 3
B : 2
C : 0
D : 0
F : 0
Appreciation Note:
Principal's Note: Exceptional performance! Keep it up!
End of Report
Ln 118, Col 8 Spaces: 4 UTF-8 CRLF Python 3.11.8 Python 3.11 (64-bit) 0-0 Go Live Previews
```

Explanation:

This section displays the student's overall academic performance by summarizing the calculated results and additional insights. It provides a clear evaluation of the student's achievements and highlights their overall progress.

- Total Marks
- Average Marks
- Highest Marks
- Lowest Marks
- Subjects Passed
- Overall Grade
- Scholarship Status
- Grade Distribution
- Appreciation Note

Multiple Student Option



```

121 |         print("", end="")
122 |     print()
123 |
124 |     # Header section
125 |     print(" ")
126 |     print("STUDENT GRADE REPORT")
127 |     print(" ")

```

Subjects Passed : 5
 Overall Grade : A
 Scholarship Status : Eligible

Grade Distribution:
 A : 3
 B : 2
 C : 0
 D : 0
 F : 0

Appreciation Note:
 Principal's Note: Exceptional performance! Keep it up!
 End of Report

Would you like to add another student? (yes/no): no

Thank you for using the Student Grade Calculator!
 PS C:\Users\HP\Desktop\Assignments\Assignment1_pythons

Explanation:

The application uses a while loop to allow users to calculate grades for multiple students in a

single execution. When the user selects "no", the application terminates gracefully with a thank-you message.

7. Learning Outcomes

- Practiced modular programming
- Applied exception handling
- Improved input validation techniques
- Used functions for better code organization
- Strengthened problem-solving skills
- Learned to generate formatted console reports

8. Conclusion

This project strengthened my understanding of Python fundamentals by combining concepts such as functions, loops, conditional statements, exception handling, lists, and formatted output into a single real-world application. It also improved my ability to write modular, maintainable, and user-friendly programs.